

Md Samshad Rahman (MACS | M.Sc. Stats | B.Sc. CS)

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Professional summary:

Data Analyst with dual Master's degrees in Applied Statistics and Computer Science. I've spent the last few years building data pipelines, cleaning messy datasets, and turning raw numbers into dashboards that people actually use to make decisions. I'm comfortable with the full process, from writing Python scripts to fix broken reporting workflows to presenting findings to non-technical audiences. I care about working somewhere where the data actually means something, where a cleaner pipeline or a sharper insight has a real effect on what the organization can do.

Relevant Projects & Experiences:

Dalhousie University, Halifax, Nova Scotia

AI-Powered Mindful Eating App (Persuasive Computing Lab)

[January 2025 – April 2025]

- Fine-tuned Llama 3.2 (3B) and deployed it on T4 GPUs, tuning memory and model parameters to hit the right balance of speed, accuracy, and cost.
- Built a safety layer to catch model hallucinations and validated outputs against 1,500 expert-reviewed inputs before they reached users.
- Built the full stack connecting a React Native app to a FastAPI backend, handling text and photo inputs in real-time so users could log meals and get instant personalized feedback.
- Integrated the Symanto NLP API to detect Big Five personality traits from user text, so the app gave advice that actually matched how each person thinks, not the same generic response for everyone.

Awarded Public Tenders Analysis Dashboard (Visual Analytics)

[September 2024 – December 2024]

- Fixed a broken NS government reporting process by writing Python scripts to ingest, clean, and standardize 10 years of raw Nova Scotia Open Data tender records.
- Wrote validation scripts that caught formatting errors and cleaned up ~35% of the noise.
- Built a custom intelligence tool using Dash/Plotly that clustered unstructured contract descriptions into coherent themes, cutting research time by ~40%.
- Established clear data dictionaries and documentation to ensure future maintainability of the pipeline.

Systems Analyst (Data Integrity & Billing), Munshi Engineers, Bangladesh

[February 2021 – April 2023]

- Managed billing infrastructure for two national power utilities (DPDC, BPDB), keeping high-volume financial data accurate and audit-ready.
- Engineered Python/SQL health checks to monitor consumption trends and flag statistical anomalies in billing data before invoice generation.
- Ran strict validation checks on every export/import cycle which maintained 100% accuracy on sensitive utility records.
- Transformed raw operational logs into actionable monthly reports using Excel and SQL, enabling management to spot revenue leakage.

Python Developer, Smartbytes Ltd. Bangladesh

[February 2022 – September 2022]

Resume Matching with Job Descriptions (German Job Market)

- Automated resume screening by building an NLP engine that parsed unstructured resumes into structured PostgreSQL fields.
- Configured the API to auto-generate documentation (Swagger), reducing integration time for the frontend team.
- Benchmarked multiple classification models to select LinearSVC, boosting data matching accuracy to 92%.

Research Assistant, Cognitive & Behavioral Data Science Lab, UI University, Bangladesh [October 2021 – March 2022]

Eating Preference Analysis Based on Human Personality Traits – Big5 (Data Science)

- Built a data ingestion pipeline for Twitter/Foursquare that respected rate limits and robots.txt rules to keep data flowing without interruptions.
- Wrote sanitization scripts to strip noise like emojis and broken URLs from thousands of raw posts and shape them into a clean, analysis-ready dataset.
- Built an NLP pipeline with spaCy and IBM Watson to extract Big Five personality traits, cutting dataset noise by ~45% and improving model accuracy by ~30%.

Software Engineer (Revenue Operations), Free Pixel Games Limited, Bangladesh

[November 2020 – May 2021]

Unified Ad-Service Revenue Dashboard

- Standardized revenue data from 5+ ad networks into one MySQL warehouse.
- Automated the daily revenue reconciliation workflow, reducing manual data consolidation time by >80% (from 2 hours to 15 mins).
- Built real-time alerts that fired whenever eCPM dropped more than 10%, catching revenue issues before they snowballed.

Additional Experience:

Microsoft Senior Advisor, Best Buy Canada Ltd. Halifax, NS

[October 2023 – Present]

- Put together a repeatable training approach for Microsoft 365 products so new team members got up to speed faster and gave customers consistent advice.
- Consistently outperformed the store's revenue-per-hour benchmark by ~35% by actually listening to what customers needed before recommending anything.
- Contributed ~29% of total departmental revenue by matching customers to the right products based on their actual technical needs.

Education:

- **Master of Applied Computer Science** [September 2023 – May 2025]
Dalhousie University, Halifax, NS
- **Master's in Applied Statistics and Data Science** [November 2019 – April 2022]
Jahangirnagar University, Bangladesh
- **BSc. in Computer Science and Software Engineering** [January 2015 – January 2020]
American International University-Bangladesh

Core competencies:

- Data Quality & Governance: Python (Pandas/NumPy), SQL (PostgreSQL/MySQL), ETL Pipelines, Data Cleansing, Root Cause Analysis, Automated Validation Scripts.
- Reporting & Visualization: Tableau, Looker Studio, Dash/Plotly, Advanced Excel (Pivot Tables, VLOOKUP), JIRA, Confluence.
- Infrastructure & Tools: AWS (Lambda, S3), Docker, Git, REST APIs, MS Office 365.